

# Jesse Bruce.

SENIOR SOFTWARE ENGINEER — PLATFORM, FULL-STACK & AI TOOLING

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► Surprise, Arizona

Senior software engineer, 8+ years across C#/.NET, TypeScript, C/C++, and Python. I modernize distributed platforms end to end (installer, datastore, messaging, secrets) and build the Claude Code / MCP developer tooling that ships product. Earlier work spanned NLP pipelines and customer-facing APIs.

## EXPERIENCE

### Senior Software Engineer, Platform & Full-Stack

Feb 2021 - Present

**FaceFirst, Inc.** (acquired by Gatekeeper Systems) · Encino, CA

C# · .NET · C++ · React · TypeScript · PostgreSQL · NATS JetStream

- Own the deployment, platform & data layer of a distributed on-prem core/edge system; senior reviewer (~90% merge rate).
- Rebuilt the server installer into a resumable wizard: 2.5 GB → 70 MB, service packages delivered OTA, edge pre-staging.
- Built the team's org-wide Claude Code plugin marketplace (TypeScript): Jira, Confluence, and Playwright MCP servers, ~40 tools, ~25 skills; org standards enforced by session and commit hooks.
- Delivered these overhauls through the AI workflow I built, under my review; the AI-assisted installer is at customer sites.
- Drove the MongoDB → PostgreSQL 17 migration across installer and platform; unlocked Ubuntu/RHEL installs.
- Moved messaging and status onto NATS JetStream, replacing drift-prone status tables with health-message reporting.
- Overhauled secrets: machine-bound Windows DPAPI replaced with a portable single-key model; zero plaintext credentials.
- Modernized the admin SPA (React 16 → 19, MUI 3 → 7, CRA → Vite); dev startup 15 s → ~430 ms.
- Initiated C++ vision-core modernization (C++14 → 17, ONNX Runtime, Linux GPU detection); moved SVN → GitHub, CI/CD.
- Currently: containerizing the platform into Linux services with per-service scaling, toward cloud deployment.

### Software Engineer II

Jun 2018 - Feb 2021

**Economic Modeling Specialists Intl. (Emsi)** · Moscow, ID — *founding member, Classifiers & Extractors team*

D (Dlang) · Python · C · Java · AWS · Terraform · SQLite

- Implemented an NLP classification & disambiguation model from academic literature; grew the founding team to 3 full-time.
- Authored APIs for internal teams and external customers across free and SaaS offerings.
- Led a team of 5 curators building a 50,000+ item training/test set; built curation tooling and dataset pipelines.
- Drove two major accuracy and load-time overhauls, profiling with perf and valgrind.

## TECHNICAL SKILLS

### LANGUAGES

C# · C++ · C · Python · TypeScript · JavaScript · D (Dlang)

### DATA & ML

PostgreSQL · MongoDB · ONNX Runtime · NLP classifiers · SQLite

### AI & DEV TOOLING

Claude Code · MCP servers · plugin & skill authoring · CI/CD

### BACKEND & WEB

.NET / ASP.NET Core · Node.js · REST APIs · React · Vite

### CLOUD & INFRA

NATS JetStream · Docker · AWS · Azure Functions · Terraform · Linux

### PROFILING & PERF

perf · valgrind · throughput analysis

## EDUCATION

### B.S., Computer Science

Washington State University · Pullman, WA

Fall 2020